THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The angle of incidence is always equal to the angle of reflection. This is the 1

(a) second law of reflection **(b)** first law of reflection
(c) Snell's law **(d)** law of refraction

2. The refractive index of a medium is the ratio of the speed of light 1

(a) in the medium to that in vacuum **(b)** in glass to that in water only
(c) in vacuum to that in the medium **(d)** in air to that in water only

3. A convex lens is also called 1

(a) a cylindrical lens **(b)** a diverging lens
(c) a converging lens **(d)** a plane lens

4. The image formed by a plane mirror is 1

(a) real and diminished
(b) virtual and magnified
(c) virtual, erect and of the same size as the object
(d) real and inverted

5. When light goes from air into glass, it 1

(a) bends towards the normal **(b)** gets reflected completely
(c) does not bend **(d)** bends away from the normal

6. The power of a lens of focal length 50 cm is 1

(a) +2 D **(b)** +0.5 D **(c)** +5 D **(d)** -2 D

7. The centre of the reflecting surface of a spherical mirror is the 1

(a) aperture **(b)** principal focus
(c) pole **(d)** centre of curvature

8. When light passes from a denser to a rarer medium, it 1

(a) bends away from the normal **(b)** goes straight
(c) bends towards the normal **(d)** stops

9. SI unit of power of a lens is 1

(a) watt **(b)** dioptre **(c)** metre **(d)** newton

10. The focal length of a spherical mirror of radius of curvature R is 1

(a) R/2 **(b)** R/4 **(c)** 2R **(d)** R

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State the laws of reflection of light. 2

12. State the sign convention for the focal length of concave and convex mirrors. 2

13. What is the lateral displacement? 2

14. Define the principal focus of a concave mirror. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. List the characteristics of the image formed by a plane mirror. 3

16. A convex lens has focal length 10 cm. An object is placed 30 cm from it. Find the image distance and magnification. 3

17. What happens to the speed and wavelength of light when it passes from air to glass? 3

OR

An object is placed at a distance of 30 cm in front of a concave mirror of focal length 15 cm. Find the position and nature of the image. 3

18. An object is placed 20 cm from a concave lens of focal length 20 cm. Find the image distance. 3

OR

What does the sign of magnification tell about the image? Give the sign for real and virtual images. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the images formed by a concave mirror for different positions of the object (at infinity, beyond C, at C, between C and F, at F, between F and P) with nature and size. 5

OR

Define power of a lens and state its unit. A convex lens of focal length 25 cm and a concave lens of focal length 50 cm are in contact. Find the net power. 5

20. State the mirror formula and the sign convention. An object 4 cm tall is placed 25 cm from a concave mirror of focal length 20 cm. Find the image position, size and nature. 5

OR

What are the uses of spherical mirrors? Explain with reasons why concave mirrors are used in torches and convex mirrors in street lights. 5

END OF QUESTION PAPER — Total: 40 Marks

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